

Superstore Dataset

Scenario

Stakeholder: Sarah Jenkins, VP of Sales

Company: Superstore (National Retail Chain)

Business Context:

Superstore has experienced steady growth in total sales volume over the past few years. However, the executive team is concerned that overall profit margins are not keeping pace with sales. Sarah suspects that aggressive discounting and an unbalanced product mix might be eroding profitability. She needs a clear picture of where the company is making money and where it is losing money before the next quarterly planning meeting.

Core Business Problem:

Despite high sales volume, overall profit margins are lower than expected. The VP of Sales needs to identify the root causes of profit loss to optimize product strategy, pricing, and regional focus for the upcoming year.

Key Questions to Answer:

1. **Product Profitability:** Which product categories and sub-categories are the biggest profit drivers, and which are actively losing money?
2. **Discount Impact:** How do different discount levels affect profitability? At what point does a discount stop driving volume and start destroying profit?
3. **Regional Performance:** Which regions and states are the most profitable, and which are underperforming?
4. **Customer Segments:** Which customer segment (Consumer, Corporate, Home Office) generates the highest revenue and the highest profit?
5. **Seasonality:** Are there specific months or quarters where sales and profits peak or drop significantly?

Deliverable:

- A clean, analysis-ready dataset.
 - A series of Pivot Tables and visualizations addressing the key questions.
 - A 1-page Executive Summary with actionable recommendations for Sarah and the executive team.
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Analysis

To: Sarah Jenkins, VP of Sales

From: Yasin Islam, Data Analyst

Subject: Root cause of profit loss Analysis & Operational Recommendations

Overview

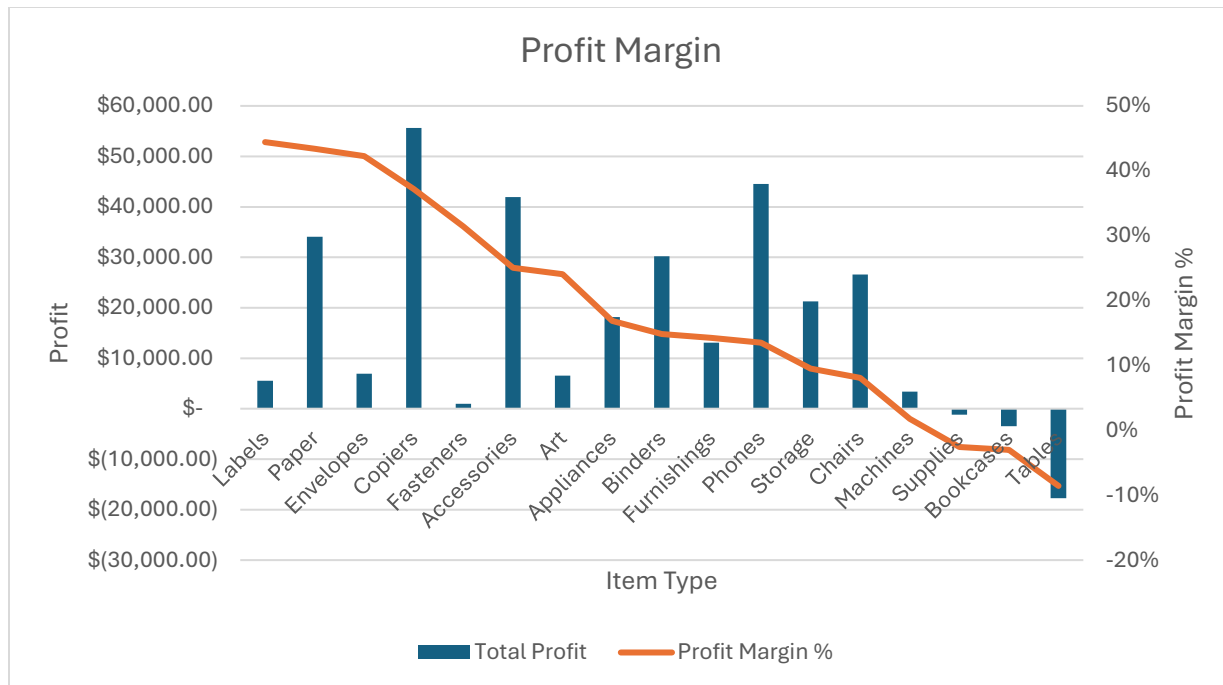
I analyzed around 10,000 purchases from the superstore dataset throughout 4 years. The provided dataset was exceptionally clean, which saved some hassle. After going through the business questions to be answered. I dealt with the data by understanding patterns and creating graphical representations for proper understanding.

Methodology

Pivot tables were created based upon business questions and alongside that, graphical representations were generated representing the pivot tables where they were custom graphs to properly understand multiple views at the same time. Certain columns were created in the pivot tables using the calculated fields to create a better picture for the data and alongside that, the data formats were fixed to ensure that they were showing proper information whilst being more readable.

Question 1 → Which product categories and sub-categories are the biggest profit drivers, and which are actively losing money?

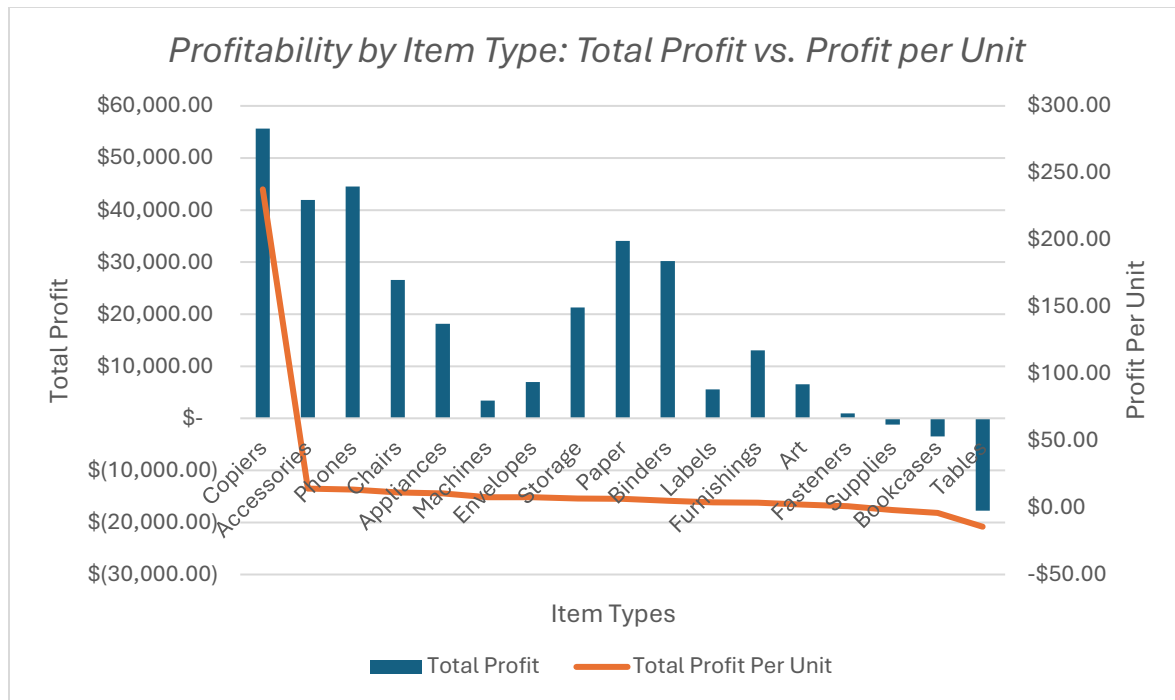
After careful observation of the data amongst the three categories: Furniture, Office Supplies and Technology – The highest profit accumulator lies within the “Technology” category with about \$145,454 whilst the least profit comes from “Furniture” Category with a total profit of \$18,451.



From the figure provided above, it is evident that the most sales we are having are phones and chairs.

chairs provides a profit of \$26,590 which is the highest from the “Furniture” Category whilst phones provide a profit of \$44,515 but the highest profit provided from “Technology” Category lies within **Copiers** which is \$55,617.

This indicates that the store would likely need to provide more focus towards the “Technology” category whilst reducing the input within “Furniture” since that is the category which is losing the store money.



Additionally, from our findings, it was even more evident that the “Furniture” category is causing more loss than profit as if you notice the provided figure above of “Profit Vs Quantities Sold” – it shows us that even though the amount of tables sold were the second lowest amongst Furniture category yet caused the highest loss of around \$17,725.

To put in the picture:

A total of 8028 quantities of furniture were sold which brought only \$18,451 profit.

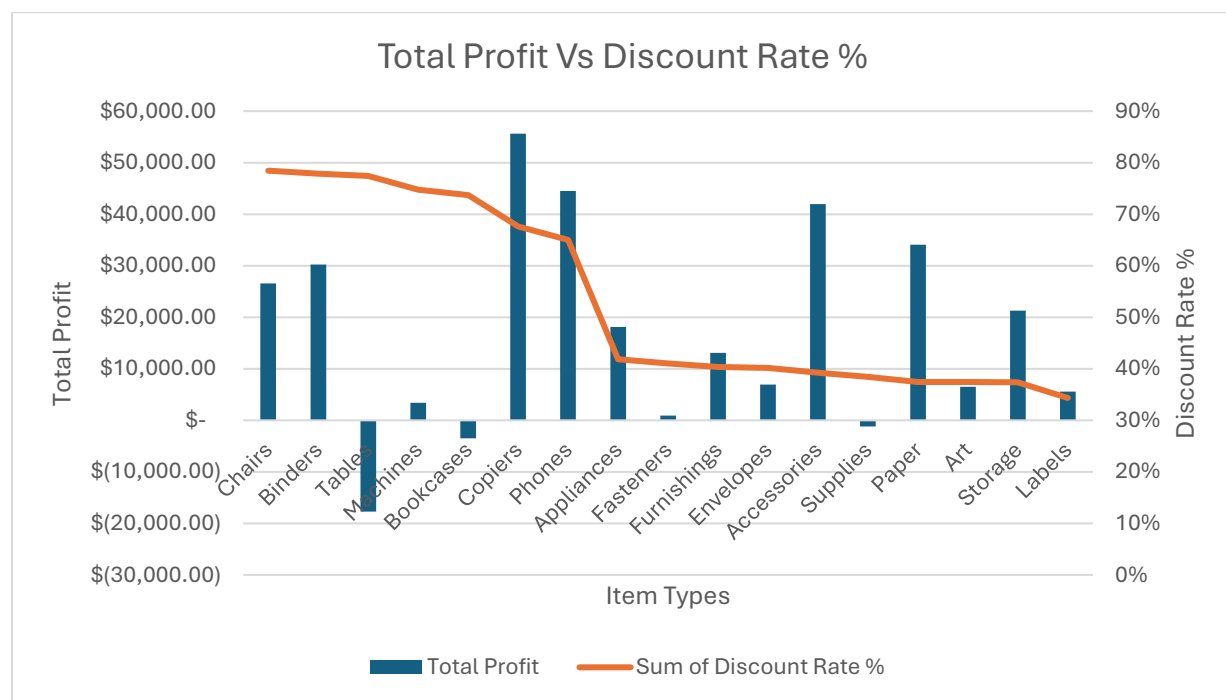
On the other hand, 6639 quantities of technology items were sold which brought \$145,454 profit.

Office supplies sold 22,906 quantities, bringing in \$122,490.

Question 2 → How do different discount levels affect profitability? At what point does a discount stop driving volume and start destroying profit?

The data proves that discounting does not universally destroy profitability. High-margin Technology items like Copiers and Phones maintain strong profitability even with discount rates exceeding 70%. However, Furniture items like Tables and Bookcases have such thin base margins that any discount above 40% pushes them into negative profitability.

Therefore, the problem is not the discount percentage itself. The problem is the **application of heavy discounts to low-margin products**. We must stop applying blanket discount rules and instead tailor discounts based on the product's base margin.

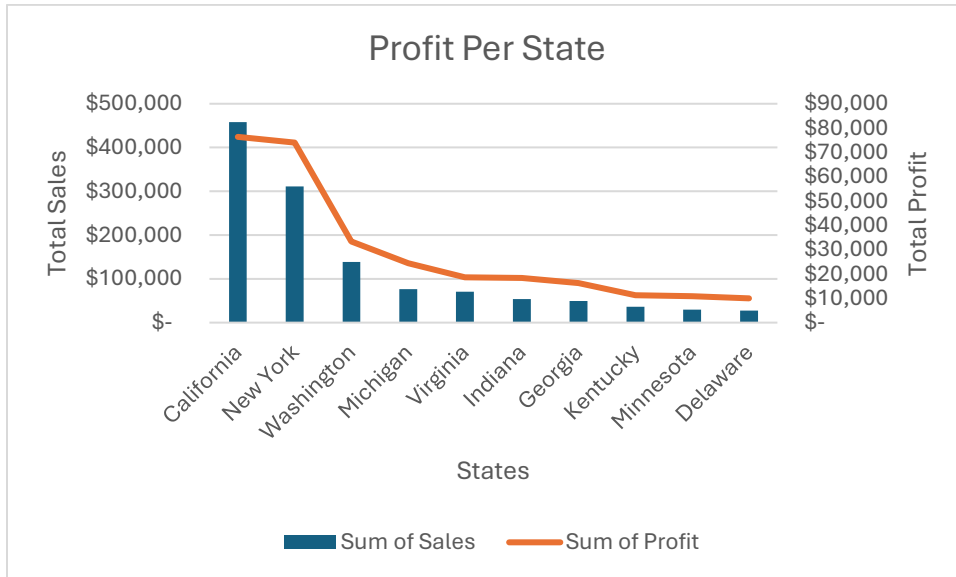


Question 3 → Which regions and states are the most profitable, and which are underperforming?

The data set provided is based in USA, showing the various sales that occurred in the different cities and its state.

After analysis, it shows that California is the state with the highest sales that have occurred alongside the highest amount of profit of around \$76,381, whereas Delaware had the least

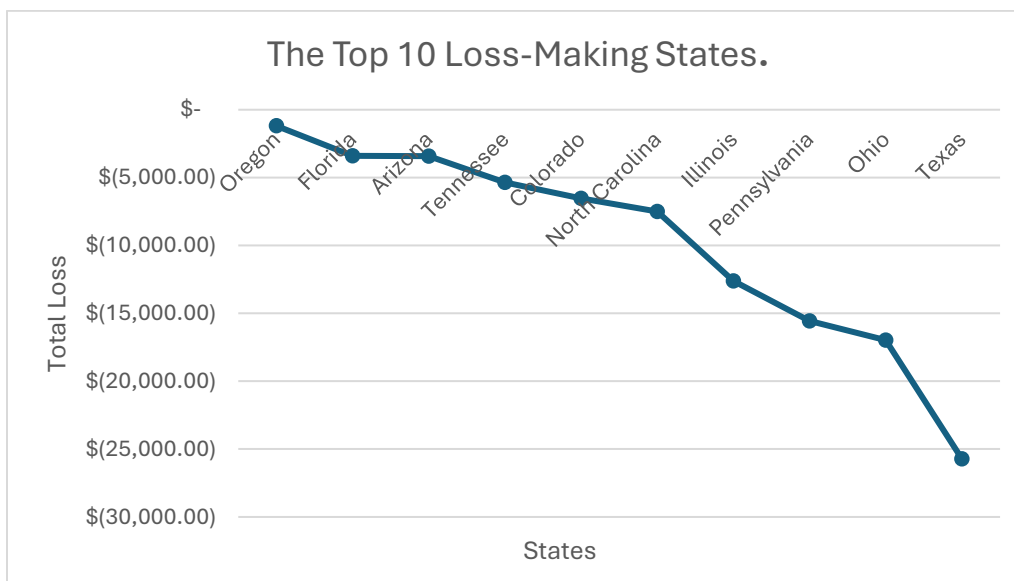
amount of sales of around \$27,451 with a profit of \$9,977.



The figure above shows the stats proving the statement.

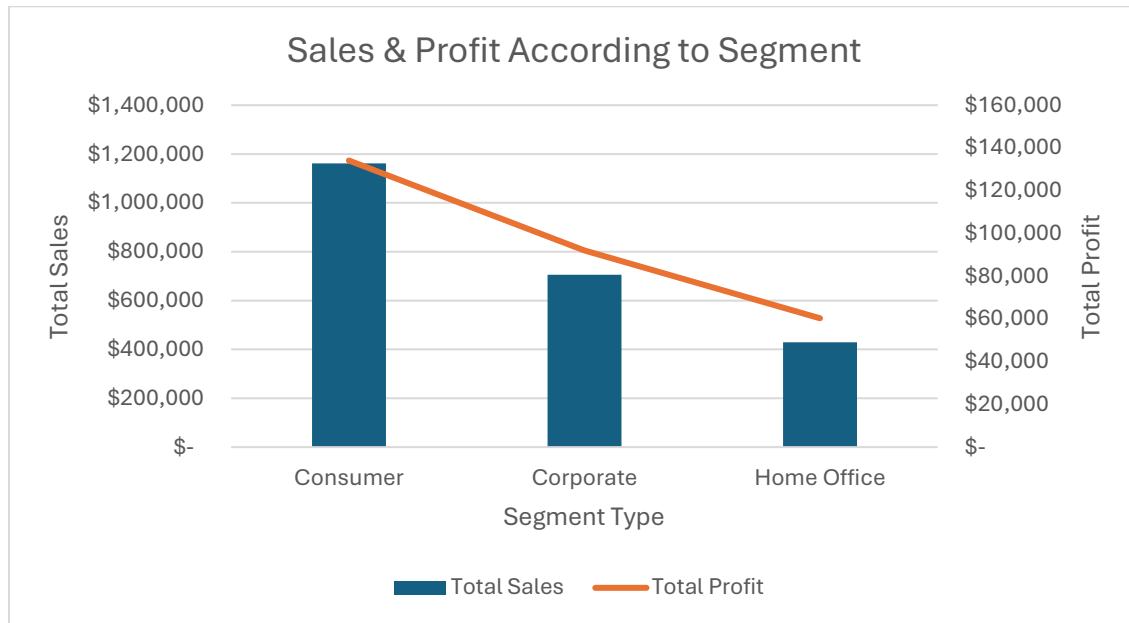
While California and New York are the most profitable states, a deeper look at the bottom 10 states by profit reveals that **Texas** is a major loss leader. Despite having high sales volume, Texas consistently loses money due to aggressive regional discounts.

Recommendation: The executive team must intervene in the Texas market to reduce discount rates and protect profit margins.



Question 4 → Which customer segment (Consumer, Corporate, Home Office) generates the highest revenue and the highest profit?

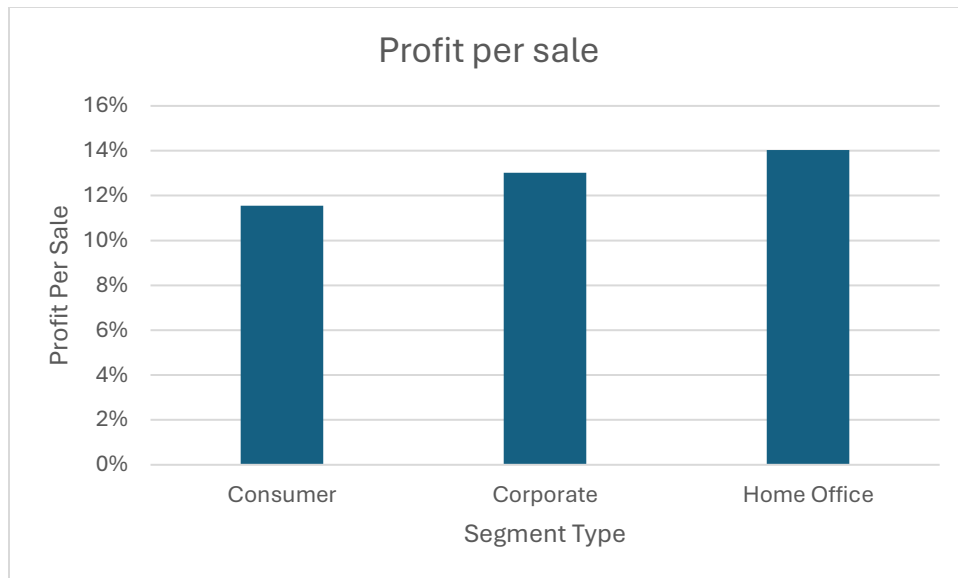
From the provided data, Customer segment brings in the most amount of revenue alongside profit. The relationship between segment profit and revenue is directly proportional. Which means that Home Office has the least amount of revenue and profit.



From the above figure we can see how the relationship between revenue and profit are directly proportional.

Therefore, Consumer has the highest revenue of \$1,161,401 whilst its profit is \$134,119 and on the other hand Home Office has a total revenue of \$429,653 and total profit of \$60,299.

Although it can be noticed that even though by numbers, Consumer brings in the most profit, but if we look at the ratio of total sales to total profit obtained, it speaks a slightly different story.

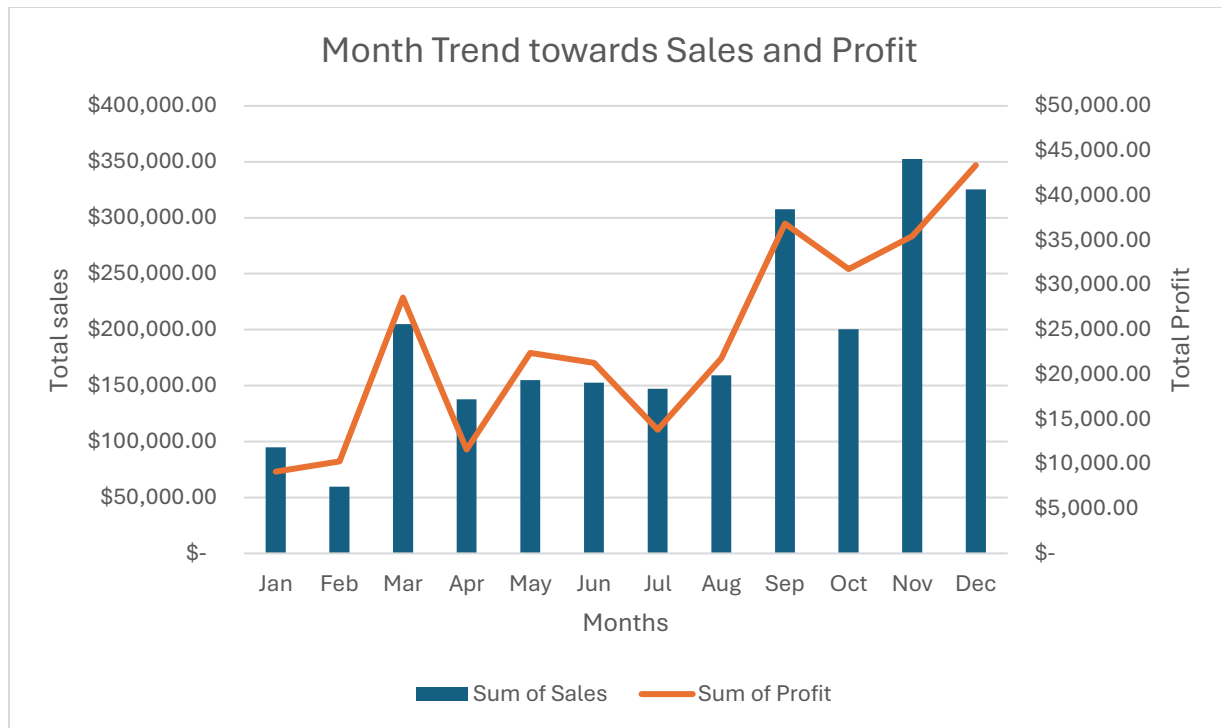


Based on the figure above, we can see that even though consumers have the higher profit, Home Office tends to have the higher profit per sale which is around 14% whilst the consumer brings in around 12%.

This means that while consumer segment is doing alright, the issue is the fact that the store is not being able to obtain a healthy ratio of profit from the total sales which means that focus needs to be provided into higher profit from consumers which could be done via the previously discussed methods of reducing discount rates.

Question 5 → Are there specific months or quarters where sales and profits peak or drop significantly?

Based on the order dates which were separated into months to understand a trend within the purchases of customers, it was found that in the following months have a surge in purchases according to the following figure.



March:

March consists of the St. Patrick's Day and a possibility of having Easter too, due to this the surge in purchase may have occurred during March.

September:

This month consists of Labor Day, which is considered a major retail weekend but of course since the items sold during this time will have discounts, hence the profit isn't higher than sales according to the diagram. And during early August to Early September, there is a back-to-school phase which causes a surge in purchase in Office Supplies and Technology and based on the earlier analysis, we figured out that technology tends to bring in a lot of sales.

October:

Consists of Halloween where the purchases of candies increase; therefore, providing the small bump in profit compared to the total sales.

November:

This month consists of a lot of sales due to major holidays like Thanksgiving, Black Friday – although, the heavy discount during this period results in the drop in profit slightly and also the Cyber Monday where there is a surge of online sales towards Technology and office supplies.

December:

Finally, we have December, consisting of Christmas which is the biggest gift giving season – therefore, consists of a big number of sales although also provided with a decent chunk of discount. Then after Christmas, there is the clearance sales resulting in more accumulated sales in this month.

When it comes to February, that's when the sales drop to the lowest although the profit still is higher than the sales, otherwise in the other months, it's mostly steady rate of sales and profits.

Conclusion

Based on the analysis of 10,000 transactions, the Superstore is experiencing a profitability crisis driven by poor discount strategies in specific product categories. To reverse this trend and optimize profit margins, I recommend the following actions:

1. **Stop Discounting Furniture Immediately:** Tables and Bookcases are losing money on every single sale due to aggressive discount (often exceeding 70%). Stop all discounts on these items. If sales volume drops, that is acceptable—we are currently losing money on every unit sold.
2. **Double Down on Technology and Home Office:** Copiers generate an incredible \$237.68 profit per unit, and the Home Office segment has the highest profit margin (14%). Shift marketing budgets to target Home Office professionals and promote high-margin Technology products like Copiers and Phones.
3. **Investigate the Texas Market:** While California and New York are our most profitable states, regional analysis reveals that certain high-volume states are underperforming. We must immediately audit the pricing and discounting strategies in loss-making states like Texas and Ohio to stop the financial bleeding.
4. **Prepare for Q4 Seasonality:** Sales peak dramatically in November and December. However, holiday discounting suppresses profit margins. We must optimize our Q4 inventory to focus on high-margin Technology bundles rather than relying on deep discounts to drive volume.